Ishaanth R. Guduru

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Objective

Seeking a Summer 2027 internship in electrical/hardware engineering. Looking to combine hands-on hardware experience — including independently designing and building an analog distortion pedal circuit — with a solid foundation in embedded systems from a differential-drive robotics project, aiming to contribute to and grow within a hardware engineering team.

Education

Georgia Institute of Technology | Atlanta, GA *May 2029*
Bachelor of Science in Electrical Engineering GPA: 4.0/4.0
Organizations: Georgia Tech Motorsports, Institute of Electrical and Electronics Engineers (IEEE), Club Pickleball

Smithtown High School West |Smithtown, New York *June 2026*
Advanced Regents Diploma GPA: 4.59/4.40

Experience

TTM Technologies | Farmingdale, NY *July – July 2025*
High School Engineering Intern
TTM Technologies is an advanced Printed Circuit Board (PCB) manufacturer and a leading supplier in technology solutions.

- Collaborated with a 15-person cross-disciplinary engineering team on PCB design and manufacturing, contributing to 2 professional projects involving microelectronics and testing radio frequency components.
- Took part in 3 product demonstrations and project construction and organization.

FRC Robotics Team 810 | Smithtown, NY *September 2022 – May 2026*
New Member Education & Youth Outreach
Team 810 is the high school robotics team for Smithtown High School West where they build multi-system robots worth 5 figures.

- Led onboarding for 45+ new FRC members annually, training them on CAD and control software; several went on to secure engineering internships.
- Mentored 3+ FLL teams and 2 FTC teams helping younger students build their experience in preparation for FIRST Robotics.

Projects

Analog Guitar Distortion Pedal *July - September 2025*
Built a working partial replica of the Boss DS-1 analog distortion pedal to develop core analog circuit design skills ahead of designing original pedal circuits.

- Prototyped the Boss DS-1's op-amp gain and diode clipping stages on a breadboard, then migrated the finalized design to a custom soldered PCB, using a multimeter to verify voltage levels and continuity at each stage during assembly.
- Modified the stock circuit by integrating a basic footswitch for on/off control and simplifying portions of the circuit.

Autonomous Course-Navigating Robot *November 2025 - March 2026*
Designed and built a differential-drive robot with mechanisms to autonomously navigate a course and manipulate objects on route.

- Implemented a PID control loop using encoder and gyroscope feedback for closed-loop navigation, tuning constants to correct for drift and improve precision.
- Integrated a gripper/claw mechanism with the Arduino Nano 33 BLE control system to enable autonomous object handling, and used an OLED display for real-time debugging feedback during development.

Leadership & Activities

Science Olympiad | President *September 2024 – June 2026*

- Coached 30+ new members on event-specific design and build techniques, preparing teams for state-level competitions.
- Led team to 3 New York State Science Olympiad competitions for the first time in school history by restructuring club schedules and practices while leading the team with multiple top 5 finishes across 4 events yearly.

Skills

Programming: Java (fluent), Python (fluent), C++ (fluent)
Hardware: Arduino Nano 33 BLE, Breadboard Prototyping, PCB Soldering, Multimeter, Encoders, Gyro Sensors, OLED Displays
Software: KiCad, Altium Designer, Arduino IDE, Autodesk Fusion 360, SolidWorks, Onshape
Languages: English (native), Telugu (native), Hindi (fluent), Spanish (limited)